Cost-aware Eisenstein Q-seed schedule

Question. Can a bounded search choose a lower source-formula cost graph for the 21 Q seeds used by one restart, while preserving the existing scalar recoding and exact native outputs?

Instance. The study uses the existing secp256k1 Eisenstein coefficient graph and frozen one-restart input in JOINT_SEED_GRAPH.md. It changes the schedule for constructing requested Q points. It does not change the curve, scalar recoding, target seed set, or online restart evaluation.

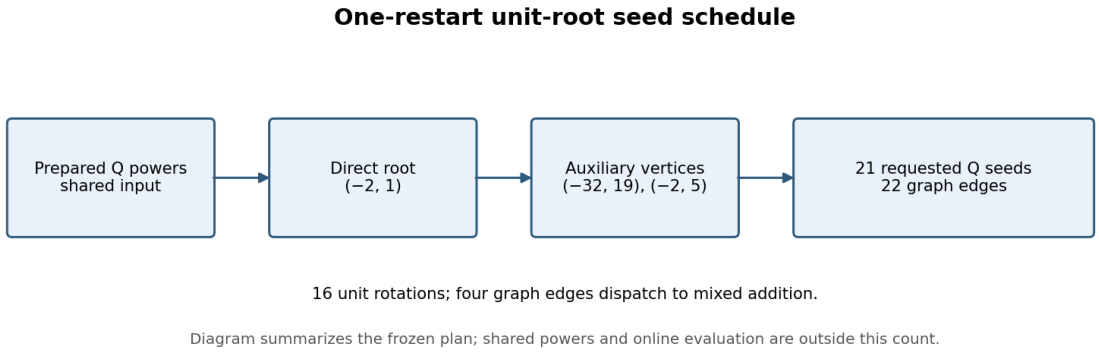


Figure 1: Cost-aware Q-seed schedule

Selected plan. The cost-aware one-restart plan starts from unit root `(-2,1)`, reached by one unit rotation of the prepared Q power. It uses auxiliary vertices `(-32,19)` and `(-2,5)` to connect 21 requested seeds through 22 graph edges. The search receipt records the source plan and output plan hashes, 10,692 second-link candidates, 372 complete graphs evaluated, source-formula boundary, and ten exact output replays. The bounded search ranks its enumerated candidates; it does not prove a global addition-chain optimum.

Modeled operation counts. The selected plan has 390 field multiplications and 105 squares, compared with 394/106 for the earlier unit-root plan, 422/113 for the one-auxiliary plan, and 437/117 for the earlier refined plan. These counts use the pinned source-formula model. They include graph construction and batch normalization, including direct-root calls, while excluding shared binary-power preparation, its inversion, recoding, plan loading, Q parsing, search, and restart evaluation. The editable count plot shows both modeled quantities and makes no wall-time inference.

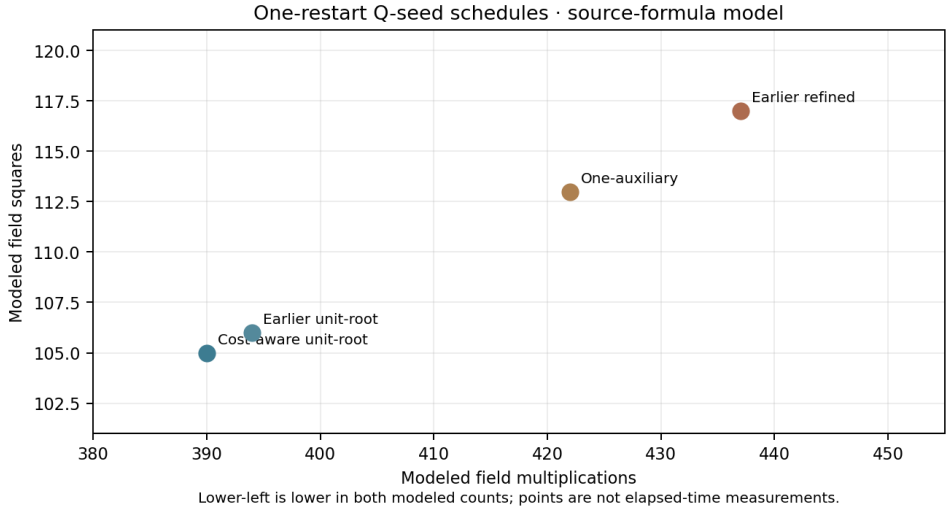


Figure 2: Modeled source-formula operation counts

Native replay and proof boundary. The existing replay receipt records ten verified exact outputs from both `graph-planned` and `fused-graph-planned` evaluators: the frozen Q plus four independently constructed alternate Q values. Its binary and fixture hashes are preserved in the receipt. The Lean graph proof proves coefficient-to-point implications under its stated additive and unit-action hypotheses. Native finite replays check these outputs only; they do not prove all-input refinement of the concrete secp256k1 group implementation.

Unmeasured performance gate. There is no qualified paired Tier 1 wall-time panel for this final schedule. Operation counts, planner receipts, and local exact-output replays do not establish lower elapsed time. The next performance measurement must pair the frozen graph orders on the same Q and scalar stream, with calibration, noise, and correctness gates; no latency or throughput gain is claimed here.

The study record links the plans, search receipts, and local replay artifacts. The editable figure source, SVG and 180-dpi PNG files, and this report's PDF accompany the result.